



United International University

School of Science and Engineering

Final Examination; Year 2025; Trimester: Spring

Course: PHY 105/2105; Title: Physics; Sec: All

Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Questions no 1, 2, 3,4 are mandatory to answer. Answer any one from question no 5 and 6.

1. (a) Two positive charges of same magnitude are placed at a distance. Why any test charge will experience no force at the middle of the two charges? Draw appropriate diagram with electric field lines. 2 CO1
(b) A dipole is placed in an electric field such that the dipole moment becomes perpendicular to the direction of electric field. Why it will experience torque? Draw appropriate diagram. 2 CO1
(c) Robert told Eduard that a ball has a charge $2 \times 10^{-16} \text{ C}$ while Eduard replied that the charge is not quantized. Who is correct and why? 2 CO1
2. (a) Charges of magnitude $100 \mu\text{C}$ each are located in vacuum at the corners A, B and C of an equilateral triangle measuring 4 meters on each side. If the charge at A and C are positive and the charge B negative, what is the magnitude and direction of the total force on the charge at B? 3 CO3
(b) Positive charges of $2 \mu\text{C}$ and $8 \mu\text{C}$ are placed 15cm apart. At what distance from the smaller charge will be the Coulomb force due to them be zero? Suppose the test charge inserted in between them is $+q_0$. 3 CO3
(c) Consider the two protons are separated at a distance 5 nm from each other. Compare the electrostatic force and gravitational force between them. The gravitational constant is $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ and the mass of each proton is $1.67262192 \times 10^{-27} \text{ kg}$. 2 CO3
3. (a) Calculate the electric field at point P from figure 01. 3 CO3

Here, $q_1 = q_3 = +5 \mu\text{C}$, $q_2 = -8 \mu\text{C}$, $r = 2 \text{ cm}$ and $R = 3 \text{ cm}$.

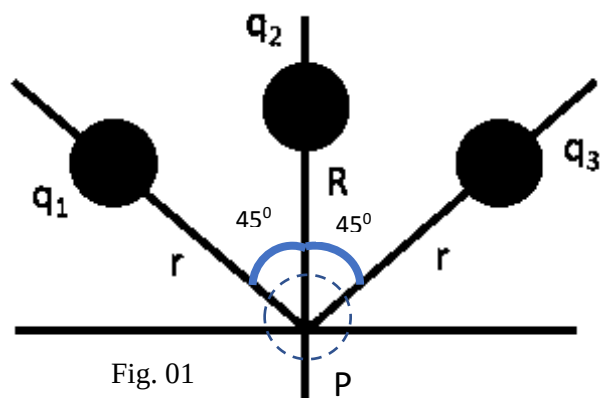


Fig. 01

- (b) A dipole has charge $q = 5 \mu\text{C}$ and the distance of the charges in the dipole is $d = 10 \text{ nm}$. 3 CO3
 - (i) If it is placed in an electric field with value $E = 100 \text{ N/C}$ and the direction of dipole moment is perpendicular to the electric field, calculate the torque acting on the dipole.
 - (ii) How much work must an external agent do to rotate the dipole from 40° to 110° in the electric field?

(c) Calculate the electric field at point P from figure 02. Here, $q_1=q_2= 5\mu\text{C}$; $q_3= -8\mu\text{C}$ and $a = 2 \text{ mm}$. 3 CO3

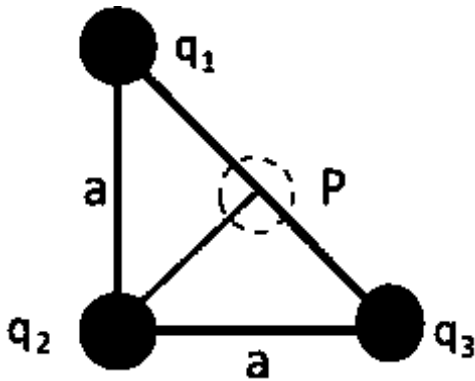


Fig. 02

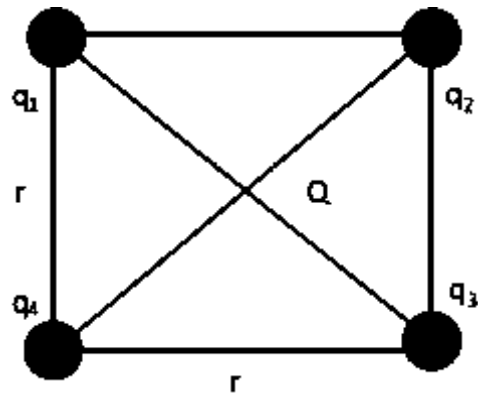


Fig. 03

4. (a) Figure 03 shows 4 charges on the four corner of a square. Here, $q_1=+5\text{C}$, $q_2= +7\text{C}$, $q_3= -8 \text{ C}$, $q_4= -10 \text{ C}$ and $r = 2 \text{ m}$. Calculate the net electric potential at the center of the square (at point Q). 3 CO3

(b) The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47 D , where $1\text{D}= 1 \text{ Debye unit} = 3.34 \times 10^{-30} \text{ Cm}$. Calculate the electric potential due to an ammonia molecule at a point 48 nm away along the axis of the dipole. 3 CO3

(c) A charge of $-2 \mu\text{C}$ is located on the y-axis at the coordinates $(0,1)$ while a second charge of $+4 \mu\text{C}$ is located on the x-axis at the coordinates $(3,0)$ from the origin. Determine the value of electric potential at the origin. 3

5. (a) Show that, the electric field due to a dipole is, 4 CO2

$$E = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}$$

where the symbols have their usual meanings.

- (b) Show that, the potential energy of dipole is, $U = -\vec{p} \cdot \vec{E}$ 4 CO2
Draw the diagram and discuss the conditions of maximum potential energy.

6. (a) Draw proper diagram and show that the potential difference, CO2

$$V_f - V_i = - \int_i^f \vec{E} \cdot \vec{d}$$

where the symbols have their usual meaning. 4

- (b) Show that, the electric potential due to an electric dipole, 4 CO2

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos\theta}{r^2}$$

Draw proper diagram and the symbols have their usual meaning.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.